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SYSTEM FOR SELECTIVE ROUTING OF A HEATED LIQUID TO AN AIR INTAKE DUCT

Abstract

A vehicle system includes an air intake assembly that has an inlet through which air flows, a filter downstream of the inlet, and an outlet downstream of the filter through which air from the filter flows, a liquid source, a valve, a temperature indicator and a controller. The valve is electrically actuated and arranged to selectively permit liquid flow from the liquid source to the air intake assembly. The temperature indicator provides an indication of a temperature in the air intake assembly upstream of the filter. And the controller is communicated with the temperature indicator and the valve and is operable to open the valve when the output of the temperature indicator indicates a temperature below a temperature threshold.

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Background/Summary

FIELD

[0001] The present disclosure relates to a system for selectively heating with a liquid of an air intake duct of an engine air intake assembly.

BACKGROUND

[0002] Vehicles include an air intake through which air is routed to an engine to support combustion in the engine. In cold weather, snow can be taken into or ice may develop within the air intake. The snow or ice can block or otherwise impeded air flow through the air intake assembly and reduce air flow to the engine which negatively impacts engine performance.

SUMMARY

[0003] In at least some implementations, a vehicle system includes an air intake assembly that has an inlet through which air flows, a filter downstream of the inlet, and an outlet downstream of the filter through which air from the filter flows, a liquid source, a valve, a temperature indicator and a controller. The valve is electrically actuated and arranged to selectively permit liquid flow from the liquid source to the air intake assembly. The temperature indicator provides an indication of a temperature in the air intake assembly upstream of the filter. And the controller is communicated with the temperature indicator and the valve and is operable to open the valve when the output of the temperature indicator indicates a temperature below a temperature threshold.

[0004] In at least some implementations, the valve is closed when the temperature in the air intake assembly upstream of the filter is above the temperature threshold.

[0005] In at least some implementations, the liquid source is a cooling circuit associated with an engine of the vehicle and the liquid flows from the cooling circuit to the valve.

[0006] In at least some implementations, the air intake assembly includes a housing that defines an intake duct between the inlet and the filter, and the liquid is routed through a passage arranged in heat transfer relationship with the intake duct. In at least some implementations, the passage is defined at least in part within a wall that defines at least part of the intake duct.

[0007] In at least some implementations, the passage is defined at least partially within a heat exchanger arranged within the intake duct, wherein the liquid flowing in the heat exchanger is maintained separate from air flowing through the intake duct.

[0008] In at least some implementations, a pump moves the liquid under pressure from the liquid source and to the valve.

[0009] In at least some implementations, the temperature indicator is responsive to a temperature between the inlet and the filter within a housing of the air intake assembly.

[0010] In at least some implementations, the temperature indicator is a sensor responsive to a temperature associated with an engine of the vehicle, or the temperature indicator provides information regarding an ambient temperature outside of the vehicle.

[0011] In at least some implementations, a method of selectively increasing the temperature within an air intake for an engine includes determining a temperature, comparing the temperature to a temperature threshold, and providing flow of a liquid from a liquid source to an intake duct of an air intake assembly when the determined temperature does not satisfy the temperature threshold.

[0012] In at least some implementations, providing the flow of the liquid is accomplished by opening a valve to permit the liquid to flow through the valve to the intake duct. In at least some implementations, the valve includes a solenoid, and the valve is opened by providing electricity to the valve and the valve is located at least partly in a conduit between the liquid source and the intake duct.

[0013] In at least some implementations, the flow of the liquid is provided for a predetermined period of time, and wherein the valve is closed after the predetermined period of time. In at least some implementations, the period of time varies as a function of the magnitude of a difference between the temperature and the temperature threshold.

[0014] In at least some implementations, the method includes redetermining the temperature one or

more times and terminating the flow of the liquid when the temperature is determined to be above a second temperature which may be equal to or greater than the temperature threshold.

[0015] In at least some implementations, the liquid source is a cooling circuit including a heat exchanger of the vehicle.

[0016] In at least some implementations, the liquid source is a cooling circuit and the liquid flows to and from an engine of the vehicle.

[0017] In at least some implementations, the method includes determining a temperature of the liquid and the step of providing the flow of liquid occurs when the determined temperature does not satisfy the temperature threshold and when the temperature of the liquid is above a liquid temperature threshold.

[0018] In the system and method, liquid that is heated by components of the vehicle can be routed to the intake duct of an air intake assembly to increase the temperature thereof. This may help to melt ice or snow that exists within the intake duct and may inhibit or prevent ice or snow accumulation within the air intake assembly in conditions in which ice or snow may be present. The system and method can selectively provide the heated liquid to the air intake assembly to avoid heating air in the assembly when heating is not desired. The system can be conveniently developed to take advantage of heat sources already present in the vehicle and to utilize otherwise wasted heat and energy to improve performance of the air intake assembly.

[0019] Further areas of applicability of the present disclosure will become apparent from the detailed description, claims and drawings provided hereinafter. It should be understood that the summary and detailed description, including the disclosed embodiments and drawings, are merely exemplary in nature intended for purposes of illustration only and are not intended to limit the scope of the invention, its application or use. Thus, variations that do not depart from the gist of the disclosure are intended to be within the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1 is a diagrammatic view of a vehicle including an engine and an air intake system via which air is supplied to the engine;

[0021] FIG. 2 is a diagrammatic view of a system for selectively increasing the temperature within an intake duct of the air intake assembly;

[0022] FIG. 3 is a plan view of a portion of a vehicle engine compartment;

[0023] FIG. 4 is a schematic view of a system for selectively increasing the temperature of an intake duct of an air intake assembly for a vehicle;

[0024] FIG. 5 is a perspective view of the air intake system; and

[0025] FIG. 6 is a flowchart of a method for selectively increasing the temperature within an intake duct of an air intake assembly.

DETAILED DESCRIPTION

[0026] Referring in more detail to the drawings, FIG. 1 illustrates a vehicle **10** having a combustion engine **12** in an engine compartment **14** of the vehicle body **16**. In the engine **12**, a mixture of air and fuel is ignited to generate motive power for the vehicle **10**. Fuel may be provided to the engine **12** by one or more fuel injectors and air may be routed to the engine **12** via an air intake assembly **18**. Engine coolant is circulated relative to the engine **12** and through a heat exchanger or radiator **20**. The radiator **20** is a liquid-to-air heat exchanger by which heat in the coolant is conducted to the metal housing of the radiator **20**, and the heat is radiated and transferred by convection to air surrounding the radiator **20** within the engine compartment **14**. This reduces the temperature of the coolant flowing therethrough and increases the temperature of the surrounding air.

[0027] As shown in FIGS. 3 and 5, the air intake assembly **18** may include a housing **22** received in

the engine compartment **14** and having an inlet **24** through which air flows into the housing **22**. The inlet **24** may be a port oriented to receive air flowing from the front of the vehicle **10** toward the rear of the vehicle **10**. Air that enters the inlet **24** flows through an intake chamber or duct **26** to a filter **28** located within the housing **22**, downstream of the inlet **24**. The filter **28** removes contaminants from the air flow, and in use, a flow of filtered air exits the filter **28** and is directed by a clean air chamber or clean air duct **30** to an outlet **32** of the air intake assembly **18** from which filtered air exits the housing **22**. Air that exits the housing **22** is routed to the engine **12** to support combustion in the engine **12**.

[0028] In at least some implementations, such as is shown in FIG. **4**, the vehicle **10** includes a turbocharger **34** arranged between the outlet **32** of the air intake assembly **18** and the engine **12**. The turbocharger **34** includes a compressor **36** arranged to discharge air at an increased pressure and provided an increased mass flow rate of air to the engine **12**. The compressor **36** may be coupled to a turbine **37** that is driven, for example, by a flow of exhaust gas from the engine **12**. The exhaust gases are at an elevated temperature, and air at higher temperatures is less dense and leads to reduce engine output for a given volume of air. For this reason, the vehicle **10** may include a heat exchanger **39** (e.g. an intercooler or charge air cooler), which may be a liquid-to-air (e.g. liquid coolant or water) heat exchanger, arranged to reduce the temperature of air delivered into the engine **12** for combustion. A charge air cooler **39** is shown in FIG. **4**, in the flow of air between the turbocharger **34** and the engine **12**.

[0029] To remove heat from the charge air cooler **39** and the engine **12**, they may be coupled to one or more heat exchangers, shown as the radiator **20** in FIG. **4**. The heat exchanger(s) **20** provide(s) lower temperature input coolant, that is driven by a pump **41**, to an inlet of an engine coolant circuit **43** and an inlet of a charge cooler coolant circuit **45** (where the circuits may have portions in common, and they circuits may be in parallel or series arrangement). Higher temperature output coolant flows back to the radiator **20** from outlets of the coolant circuits **43**, **45**. The temperature of the coolant is then reduced in the radiator **20** and returned to the inlets of the coolant circuits **43**, **45** before being pumped back to the coolant circuits **43**, **45**. Additionally, other liquids used in the vehicle, like engine oil and transmission fluid, may also be routed to one or more heat exchangers arranged to reduce the temperature of these fluids.

[0030] In cold weather, ice can form and snow can enter the inlet **24** of the air intake housing **22** and accumulate in the intake duct **26**. The ice or snow can restrict air flow to the filter **28** and thus, to the engine **12**, and this can starve the engine **12** of air and decrease engine performance. To inhibit or prevent ice formation, or to melt any ice or snow in the air intake assembly **18**, the vehicle **10** includes a system to increase the temperature within the intake duct **26** of the air intake assembly **18**.

[0031] In at least some implementations, as shown in FIG. **2**, the system includes a source **38** of heated liquid, and a conduit **40** communicated with the source **38** of heated liquid, and with the intake duct **26** of the air intake housing **22**. The heated liquid source **38** may be a cooling circuit for one or more powertrain components of the vehicle, for example, the engine, transmission, turbocharger, charge cooler or intercooler which use liquids and heat exchangers to control the temperature of the liquids. The liquid from the liquid source and within the cooling circuits could be a coolant (as in the cooling circuits noted herein), engine oil, transmission fluid or the like.

[0032] The air intake housing **22** may include a heat exchanger **47** or a heat exchanger **47** may be received in part of the intake duct **26**, if desired. The heat exchanger **47** may be arranged so that the heated liquid coolant increases the temperature of air within the intake duct **26**. In at least some implementations, this may be done by contact between the air and a heat conducting surface of the heat exchanger **47**, such as a metal tube or metal body of the heat exchanger **47**, or by conduction between the heat exchanger **47** and the housing **22** of the air intake assembly **18**, where the heat exchanger **47** may be located outboard of and in contact with a wall defining part of the intake duct, in an implementation. The heat exchanger **47** may be a simple tube routed through part of the

interior of the intake duct **26**, and the tube may be coiled, if desired, to increase the surface area of the tube within the intake duct **26**, or the tube may engage a surface of the housing **22** without being located within the interior, or parts of the heat exchanger **47** may be located inside and outside the intake duct, if desired. As heated liquid flows through a passage **49** (labeled in FIG. **4**) defined by the tube or tubes of the heat exchanger **47**, heat from the liquid is transferred to air flowing toward the filter in the intake duct **26** and the temperature of the air increases. In this way, the heat exchanger **47** is arranged within the flow of air between the inlet **24** and the filter chamber **28** of the air intake housing **22**, and air and snow may directly contact a surface of the heat exchanger **47** or a surface the temperature of which is altered by the flow of liquid to the heat exchanger and air intake assembly.

[0033] In at least some implementations, the heat exchanger **47** may be integrally defined by the housing **22** wherein heated liquid may be routed through a chamber or passage formed in the housing **22** which acts as a heat exchanger when liquid is route therethrough. The chamber or passage **49** is separate from the interior of the intake duct **26** so that the heated liquid does not enter the interior of the intake duct **26** or flow to the filter **28**. In such an arrangement, an inlet **51** to the chamber or passage **49** would receive the heated liquid which would flow out of the chamber or passage and back to the radiator **20** through an outlet **53**. Here, the walls or surfaces defining the interior of the intake duct **26** would be heated by the liquid in the chamber or passage **49** and this would increase the temperature within the intake duct **26**. Improved heat transfer can occur with material of the heat exchange **47** (which may be a portion of the intake duct **26** or a separate structure located at the housing **22**) having higher thermal conductivity. To permit water from melted snow or ice to exit the housing, the housing may include a drain opening **55** (FIG. **5**) and a one-way valve **57** arranged to permit water to exit from the intake duct while inhibiting or prevent air flow therethrough. The drain opening **55** may be located in a lowest area of the intake duct, relative to gravity and the orientation of the housing when installed in a vehicle.

[0034] In at least some implementations, heated liquid is provided to the intake duct **26** only some of the time and not continually. In such implementations, a valve **44** may be provided to selectively permit and prevent, or at least significantly inhibit, liquid flow from the conduit **40** and to the air intake housing **22**. The valve **44** may be electrically actuated, such as by including or being driven by a solenoid, to move between a first position and a second position, in which a greater flow of liquid is permitted than when the valve **44** is in the first position. The valve **44** may have a valve head **46** received within the conduit **40**, and in the first position the valve head **46** may mostly or fully block the flow area of the conduit **40** to inhibit or prevent liquid flow therethrough. In the second position, the valve head **46** is moved relative to the first position and a greater effective flow area is provided between the valve head **46** and the conduit **40** than when the valve head **46** is in the first position. Flow area, as used herein, is a cross-sectional area through which liquid may flow within the conduit **40** and around the valve head **46**. In the first position, the valve head **46** may reduce the effective flow area of the conduit **40** by 90% or more, including up to 100%, in at least some implementations.

[0035] In at least some implementations, the valve **44** is controlled in response to a temperature, which may be determine by a temperature sensor **48**, and the valve **44** may be opened when the temperature is below a temperature threshold, and the valve **44** may be closed when the temperature is above the temperature threshold. The temperature threshold may be set with regard to a temperature at or below which ice or snow may be likely to form at or near the air intake housing **22**. The temperature may be an ambient temperature (e.g. temperature outside the vehicle **10**), or a temperature at or near the air intake housing **22**, which may include within the intake duct **26** of the air intake housing **22**. In this regard, the temperature sensor **48** may be a sensor used to provide an indication of ambient temperature within the vehicle **10**, such as via an infotainment system of the vehicle **10**, may be obtained from a weather information source (e.g. via a telematics unit of the vehicle **10**), may be a temperature sensor **48** used by other vehicle systems (e.g. to sense

the temperature of part of the engine **12**, engine coolant or engine oil) or may be a dedicated temperature sensor used for the purpose of controlling the valve **44**. For example, a temperature sensor **48** may be mounted to the housing **22** and may have a portion exposed to air within the intake duct **26** or otherwise be arranged to be responsive to the temperature within the intake duct **26**. The temperature sensor **48** may provide an output indicative of the temperature within the intake duct **26**, which may be a direct measurement or determination of that temperature or which may be inferred from other temperatures and or time or other information relating to use of the vehicle **10**.

[0036] To control the valve **44**, the system includes a control system **50** with a controller **52** that is communicated with the temperature sensor **48** and with the valve **44**. The controller **52** is operable to open the valve **44** when the output of the temperature sensor **48** indicates a temperature below a temperature threshold, and to close, or permit the valve **44** to move to the closed position such as under the force of a biasing member like a spring, when the temperature is at or above the temperature threshold. In the example wherein the valve **44** includes a solenoid, the controller **52** may selectively provide electricity to the valve **44** to selectively drive the valve **44** and change the position of the valve **44**.

[0037] In at least some implementations, it may be desirable to increase the temperature within the intake duct **26** only in conditions in which snow or ice may exist within the intake duct **26**. In other instances, it might not be desirable to increase the temperature of the air in the intake duct **26** because air at higher temperatures is less dense and leads to reduce engine output. Thus, in general, it may be desirable to provide colder air to the engine **12** and so, in at least some implementations, increasing the temperature in the intake duct **26** by routing heated liquid to the intake duct **26** may be limited to conditions in which snow or ice may be present within the intake duct **26**. In at least some implementations, the threshold temperature is between 32 degrees Fahrenheit and 36 degrees Fahrenheit.

[0038] In order to perform the functions and desired processing set forth herein, as well as the computations therefore, the control system **50** may include, but is not limited to, one or more controller(s), processor(s), computer(s), DSP(s), memory, storage, register(s), timing, interrupt(s), communication interface(s), and input/output signal interfaces, and the like, referred to be reference numeral **52** in FIG. **2**, as well as combinations comprising at least one of the foregoing. For example, the control system **50** may include input signal processing and filtering to enable accurate sampling and conversion or acquisitions of such signals from communications interfaces and sensors. As used herein the terms control system **50** may refer to one or more processing circuits such as an application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that executes one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described functionality. The control system **50** may be distributed among different vehicle modules, such as an infotainment control module, engine control module or unit, powertrain control module, transmission control module, and the like, if desired.

[0039] The term “memory” or “storage” as used herein can include volatile memory and/or non-volatile memory, generally referred to by reference numeral **54** in FIG. **2**. Non-volatile memory can include, for example, ROM (read only memory), PROM (programmable read only memory), EPROM (erasable PROM), and EEPROM (electrically erasable PROM). Volatile memory can include, for example, RAM (random access memory), synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), and direct RAM bus RAM (DRRAM). The memory can store an operating system that controls or allocates resources of a computing device.

[0040] FIG. **6** illustrates a method **60** of selectively increasing the temperature within an air intake assembly **18** of a vehicle **10**. The method **60** may be run after the vehicle **10** is turned on (e.g. keyed on, or engine started, or the like) and in step **62**, a temperature is determined. The

temperature may be determined from information from any suitable temperature indicator, which may be a source of temperature data within or remotely located from the vehicle **10**, or a temperature sensor **48** of the vehicle **10**, for example and as noted herein.

[0041] Next, in step **64**, the determined temperature is compared to a threshold temperature. If the determined temperature is not less than the threshold temperature, then the method **60** ensures that the valve **44** is in the second position, inhibiting or preventing liquid flow from the heated liquid source to the air intake assembly **18**. If not, the method **60** may skip to step **66** and cause the valve **44** to move to the second position (or remain in the second position if the valve is already in the second position) and then the method **60** may end.

[0042] If desired, instead of ending after one iteration, the method **60** may loop back to step **62** one or more times to check or determine the temperature again and compare again the newly determined temperature to the threshold temperature in step **64**. A counter may be used and incremented in step **68** each time the determined temperature is not less than the threshold temperature and the method **60** may end after the counter reaches a desired number as checked in step **70**. In at least some implementations, the counter value needed to terminate the method **60** may be set as a function of the difference between the determined temperature and the threshold temperature with a greater differential requiring fewer iterations (down to one iteration), as the confidence that snow or ice conditions are not present is greater when the determined temperature is higher.

[0043] If in step **64** the determined temperature is less than the threshold temperature, then the method **60** continues to step **72** in which it is determined if the valve **44** is in the first position in which liquid flow is permitted by the valve **44**. If the valve **44** is not in the first position, then the method **60** proceeds to step **74** in which the valve **44** is moved to the first position (e.g. opened) to permit liquid flow, or an increased liquid flow, from the heated liquid source to the air intake assembly **18**. In at least some implementations, the method **60** may include a step in which the temperature of coolant is determined and compared to coolant temperature threshold. In this example method, the valve is then opened only when the coolant temperature is above a threshold to ensure that the liquid routed to the air intake assembly is capable of heating the air intake assembly.

[0044] In the example wherein the temperature determination is based on the output from a temperature sensor **48** that is responsive to the instantaneous or current temperature within the air intake assembly **18** (e.g. within the intake duct **26**) then the method **60** may loop back to step **62**, perhaps after expiration of a time period determined by a timer associated with the controller **52** to allow some time for the temperature in the air intake assembly **18** to increase or with use of a counter as noted above. This new, updated or more current temperature determination is then compared to the threshold temperature in step **64** or to a second temperature which may be a set value or a variable threshold, and which may be the same as or different than the original or first temperature threshold.

[0045] In other example methods, the system may assume a temperature increase in the air intake assembly **18** occurs after a certain amount of time after the engine **12** has been operating, and after this time, the method **60** may move the valve **44** to the second position and then end. This assumption of temperature increase may be made based on empirical data taken over time and related to the temperature within the engine compartment **14** of the vehicle **10** as a function of operation of the engine **12**, engine oil temperature or other engine temperature, or the like, which may be mapped or otherwise determined as a function of or independently of an ambient temperature. The time period may be varied based upon the difference between the determined temperature and the threshold temperature, with a greater time period provided when the determined temperature is lower.

[0046] For example, if the determined temperature is ten degrees Fahrenheit below the threshold temperature, then the method **60** may cause the valve **44** to be opened (in the first position) for a

predetermined time period deemed sufficient to ensure that the temperature within the intake duct **26** increases at least ten degrees before the valve **44** is closed (moved to the second position). In this example, when the determined temperature is five degrees Fahrenheit below the threshold temperature, the time period might be less. In this way, further iterations of redetermining the temperature and comparing the redetermined temperature to the threshold temperature are not needed. This may facilitate use of ambient temperature such as may be determined by a temperature sensor **48** of the vehicle **10** or a remote temperature/weather data source, or an engine temperature, to conveniently be used to determine when the valve **44** should be opened so that a dedicated temperature sensor is not needed for the air intake assembly **18** in some implementations. [0047] The systems and method herein can, among other things, prevent or inhibit or reduce ice or snow within the intake duct of the air intake assembly to improve air flow through the assembly and to the engine. The system may intelligently control a flow of heated liquid to the air intake assembly to increase the temperature within the air intake assembly only when needed and avoid unduly heating air within the air intake assembly when not needed.

Claims

1. A vehicle system, comprising: an air intake assembly including an inlet through which air flows, a filter downstream of the inlet, and an outlet downstream of the filter through which air from the filter flows; a liquid source; a valve that is electrically actuated and arranged to selectively permit liquid flow from the liquid source to the air intake assembly; a temperature indicator providing an indication of a temperature in the air intake assembly upstream of the filter; and a controller communicated with the temperature indicator and the valve and operable to open the valve when the output of the temperature indicator indicates a temperature below a temperature threshold.
2. The system of claim 1 wherein the valve is closed when the temperature in the air intake assembly upstream of the filter is above the temperature threshold.
3. The system of claim 1 wherein the liquid source is a cooling circuit associated with an engine of the vehicle and wherein the liquid flows from the cooling circuit to the valve.
4. The system of claim 1 wherein the air intake assembly includes a housing that defines an intake duct between the inlet and the filter, and wherein the liquid is routed through a passage arranged in heat transfer relationship with the intake duct.
5. The system of claim 4 wherein the passage is defined at least in part within a wall that defines at least part of the intake duct.
6. The system of claim 4 wherein the passage is defined at least partially within a heat exchanger arranged within the intake duct, wherein the liquid flowing in the heat exchanger is maintained separate from air flowing through the intake duct.
7. The system of claim 1 which also includes a pump that moves the liquid under pressure from the liquid source and to the valve.
8. The system of claim 1 wherein the temperature indicator is responsive to a temperature between the inlet and the filter within a housing of the air intake assembly.
9. The system of claim 1 wherein the temperature indicator is a sensor responsive to a temperature associated with an engine of the vehicle, or wherein the temperature indicator provides information regarding an ambient temperature outside of the vehicle.
10. A method of selectively increasing the temperature within an air intake for an engine, comprising: determining a temperature; comparing the temperature to a temperature threshold; and providing flow of a liquid from a liquid source to an intake duct of an air intake assembly when the determined temperature does not satisfy the temperature threshold.
11. The method of claim 10 wherein providing the flow of the liquid is accomplished by opening a valve to permit the liquid to flow through the valve to the intake duct.
12. The method of claim 11 wherein the valve includes a solenoid, and the valve is opened by

providing electricity to the valve and the valve is located at least partly in a conduit between the liquid source and the intake duct.

13. The method of claim 11 wherein the flow of the liquid is provided for a predetermined period of time, and wherein the valve is closed after the predetermined period of time.

14. The method of claim 13 wherein the period of time varies as a function of the magnitude of a difference between the temperature and the temperature threshold.

15. The method of claim 10 which also includes redetermining the temperature one or more times and terminating the flow of the liquid when the temperature is determined to be above a second temperature which may be equal to or greater than the temperature threshold.

16. The method of claim 10 wherein the liquid source is a cooling circuit including a heat exchanger of the vehicle.

17. The method of claim 10 wherein the liquid source is a cooling circuit and the liquid flows to and from an engine of the vehicle.

18. The method of claim 10 which also includes determining a temperature of the liquid and wherein the step of providing the flow of liquid occurs when the determined temperature does not satisfy the temperature threshold and when the temperature of the liquid is above a liquid temperature threshold.
